

# POINT PROCESSING & GRAY LEVEL TRANSFORMATIONS



## POINT PROCESSING

- Image enhancement is the process of adjusting digital images so that the results are more suitable for display or further image analysis.
- It has two broad categories:
  - ❖ Spatial domain methods
  - ❖ Frequency domain methods
- Spatial domain methods are operate directly on the pixels.
- Point processing operation deals with pixel intensity values individually.

## POINT PROCESSING (CONTD)

- The intensity values are altered using particular transformation techniques as per the requirement.
- Enhanced at any point in an image depends only on the gray level at that point techniques are referred as **point processing**.
- Most spatial domain enhancement operations can be reduced to the form of,

$$g(x, y) = T[f(x, y)]$$

- In this case T is referred to as a **gray level transformation function** or a **point processing operation**.

## POINT PROCESSING (CONTD)

- where  $f(x, y)$  is the input image,  $g(x, y)$  is the processed image and  $T$  is point operator defined over some neighborhood of  $(x, y)$ .

- Point processing operations take the form of,

$$s = T(r)$$

- where  $s$  refers to the processed image pixel value and  $r$  refers to the original image pixel value.
- Mask** is a small matrix useful for blurring, sharpening, edge detection.

## POINT PROCESSING (CONTD)

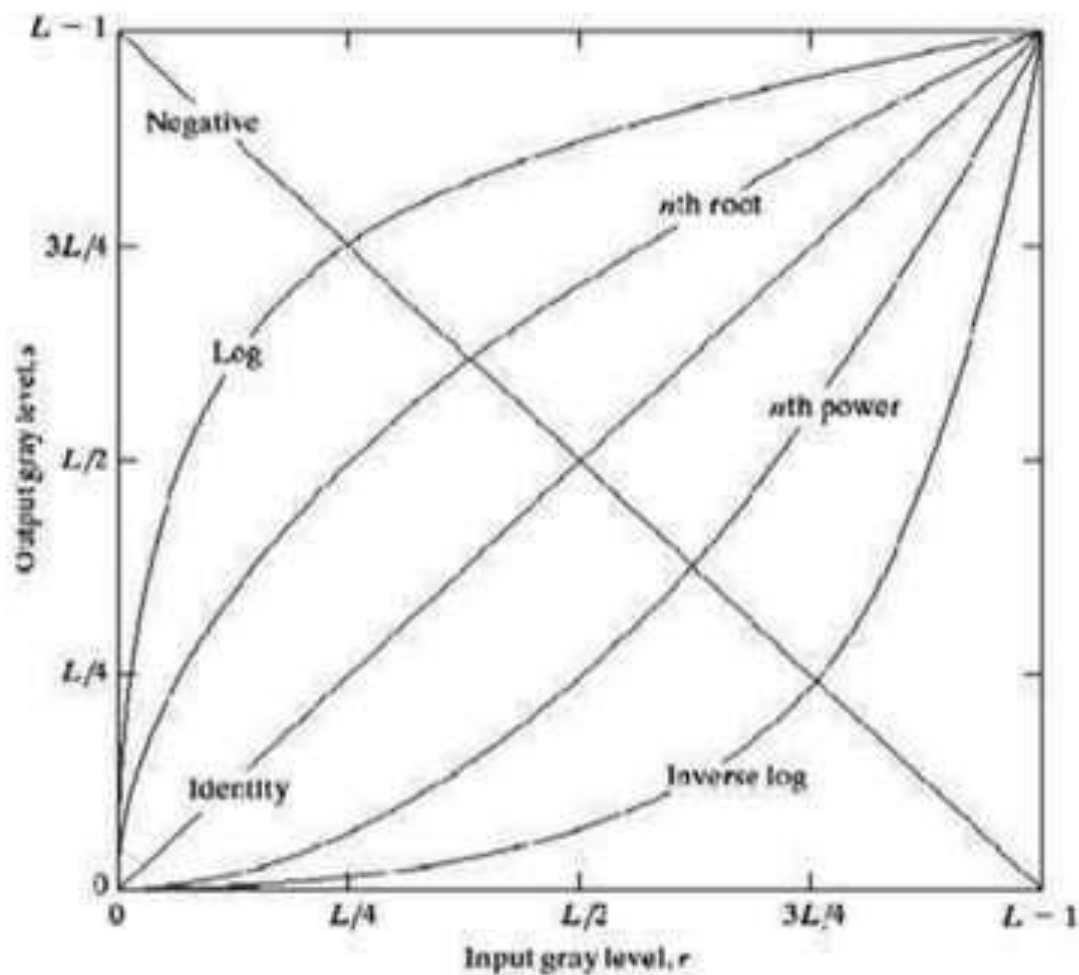
- New image is generated by multiplying the input image with the mask matrix. Mask can be in any dimension (i.e 3x3, 4x4).
- Contrast stretching expands the range of intensity levels in an image.
- Extreme contrast stretching yields Thresholding.
- Thresholding image has maximum contrast as it has only Black & White gray values.
- Brightness enhancement is shifting of intensity values to higher level.



# BASIC GRAY LEVEL TRANSFORMATION

- There are three basic gray level transformation.
  - ❖ Linear
  - ❖ Logarithmic
  - ❖ Power – law

# GRAY LEVEL TRANSFORMATION GRAPH



# LINEAR TRANSFORMATION

- Linear transformation includes following two categories,
  - ❖ Negative Transformation
  - ❖ Identity Transformation



## NEGATIVE TRANSFORMATION

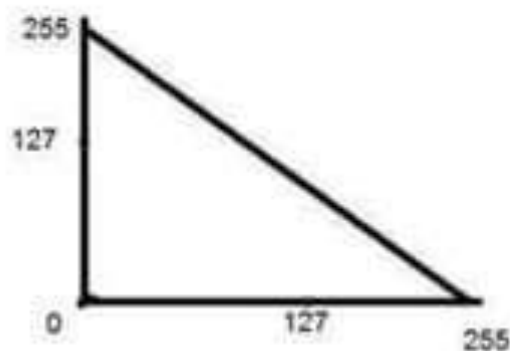
- Negative images are useful for enhancing white or grey detail embedded in dark regions of an image.
- Negative transform exchanges dark values for light values and vice versa.
- The Negative Transformations can be defined by,

$$s = (L-1-r)$$

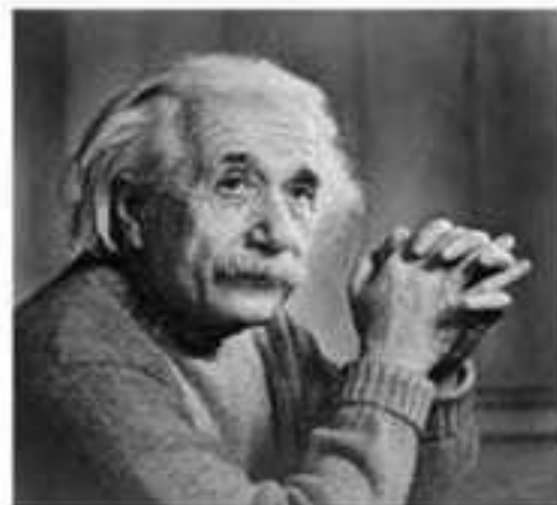
- Negative of an image intensity levels in the range  $[0, L-1]$ ,  
 $L-1$  = Maximum pixels value  
 $r$  = Pixel value of an image

# NEGATIVE TRANSFORMATION EXAMPLE

Graph representation



Input image



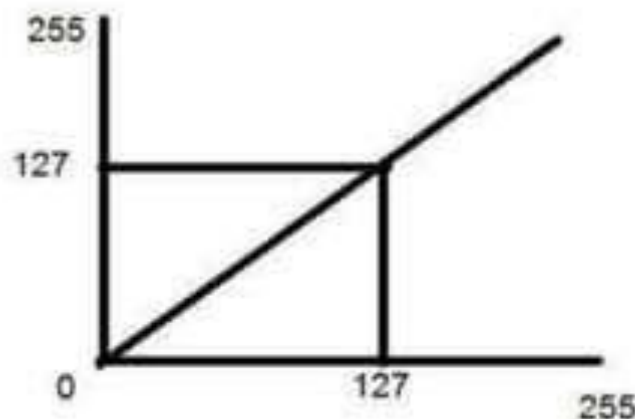
Output image



## IDENTITY TRANSFORMATION

- Each value of the input image is directly mapped to each other value of output image.
- That results in the same input image and output image.

Graph representation



## LOG TRANSFORMATION

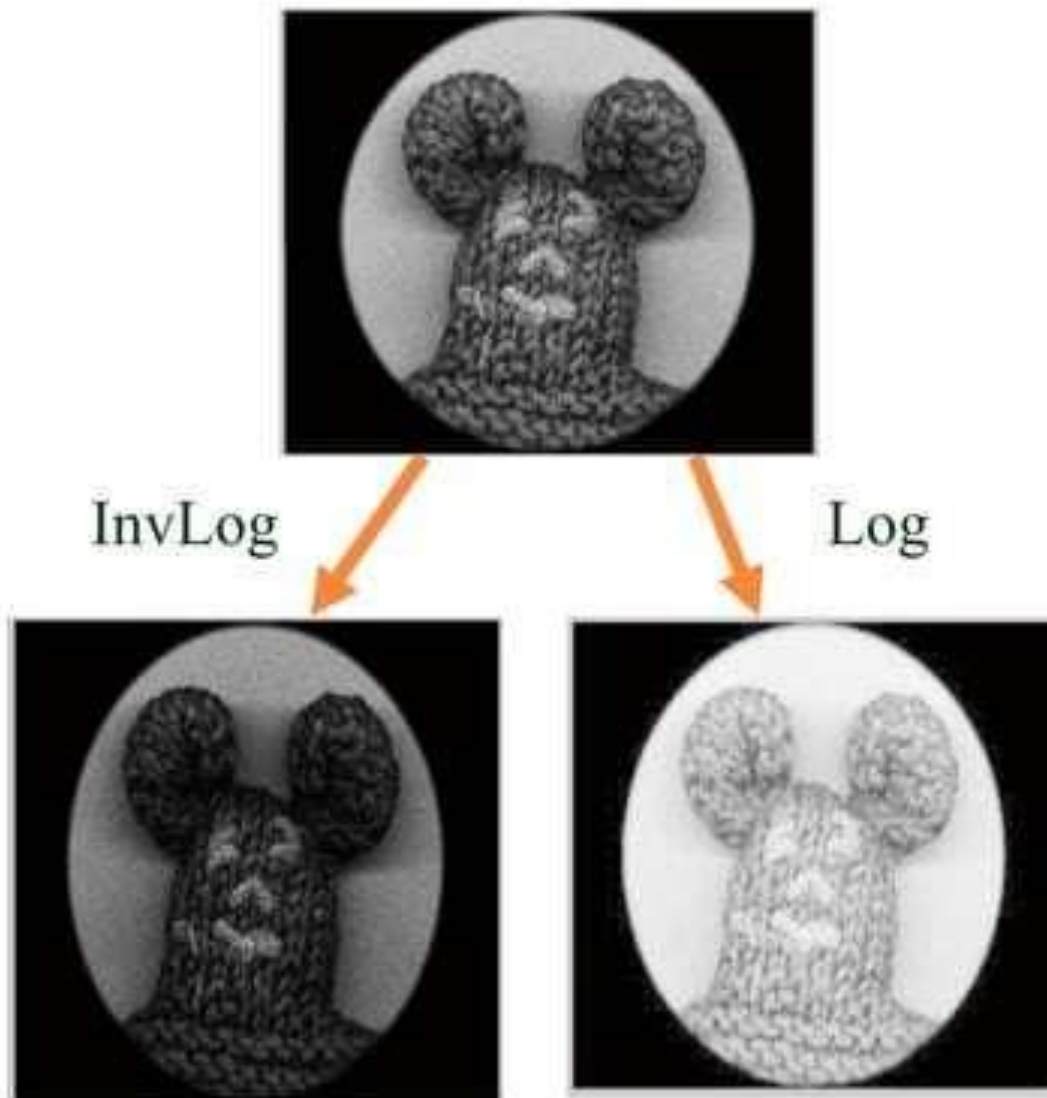
- This transform is used to expand values of dark pixels and compress values of bright pixels.
- It maps a narrow range of low level gray scale intensities into wider range of output values.
- Similarly maps the wide range of high level gray scale intensities into a narrow range of high level output values.
- The log transformations can be defined by this formula

$$s = c \log(r + 1)$$

## LOG TRANSFORMATION (CONTD)

- Where  $s$  and  $r$  are the pixel values of the output and the input image and  $c$  is a constant.
- The value 1 is added to each of the pixel value of the input image because if there is a pixel intensity of 0 in the image, then  $\log(0)$  is equal to infinity.
- So 1 is added, to make the minimum value at least 1.
- The inverse log transform is opposite to log transform.

# LOG TRANSFORMATION EXAMPLE





## POWER LAW TRANSFORMATION

- This type of transformation is used for enhancing images for different type of display devices.
- These transformations can be given by,

$$s = cr^{\gamma}$$

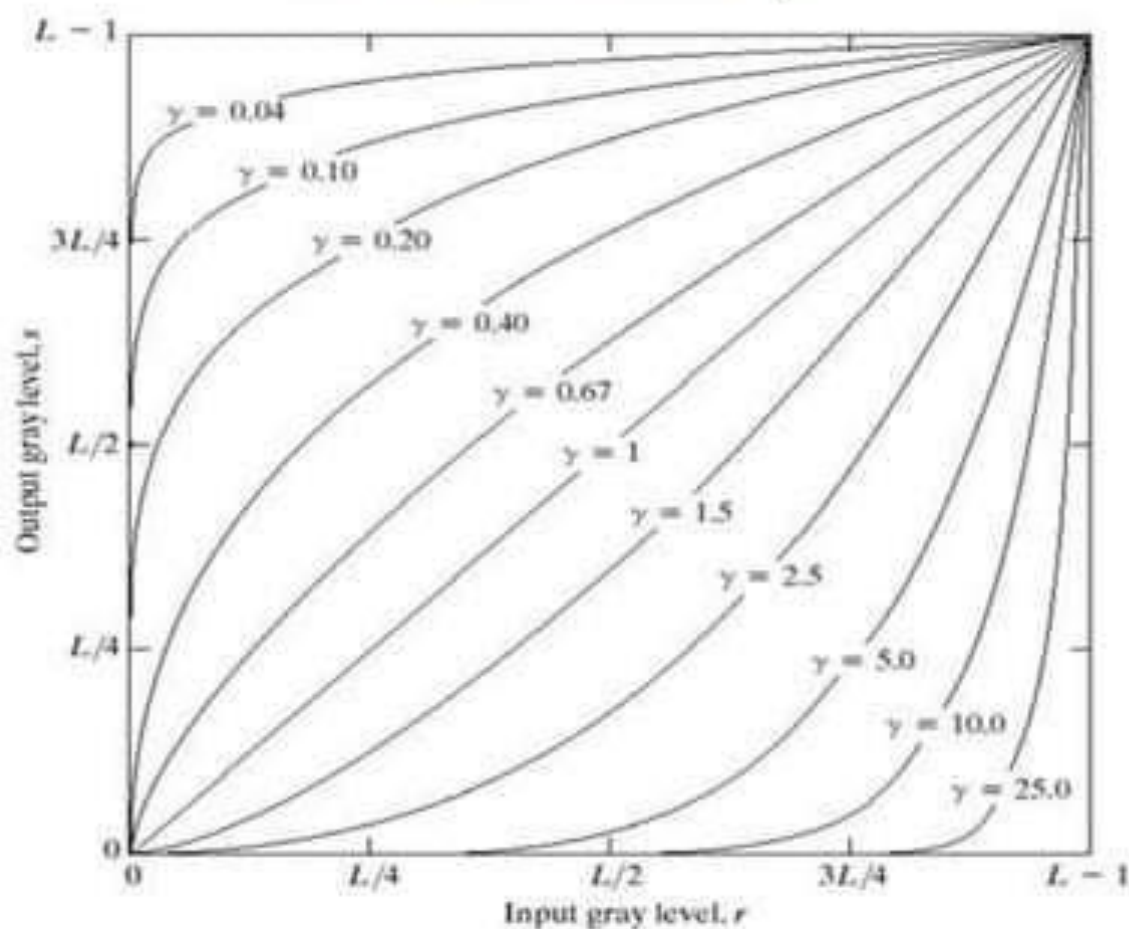
- Here, s is output pixel value, r is the input pixel value, c and  $\gamma$  are real numbers.
- Variation in the value of  $\gamma$  varies the enhancement of the images. This technique is commonly called as **Gamma correction**.

## POWER LAW TRANSFORMATION

- Different display monitors display images at different intensities and clarity because every monitor has built in gamma correction in it with certain gamma ranges.
- A good monitor automatically corrects all the images displayed on it for the best contrast to give user the best experience.
- The difference between the log transformation function and the power law functions is that using the power law function a group of possible transformation curves can be obtained just by varying  $\gamma$ .

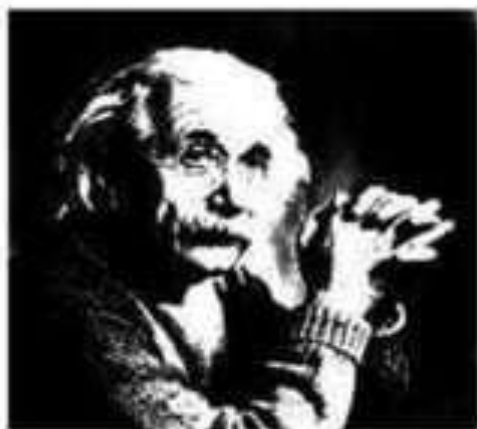
# POWER LAW TRANSFORMATION (CONTD)

Various values for  $\gamma$

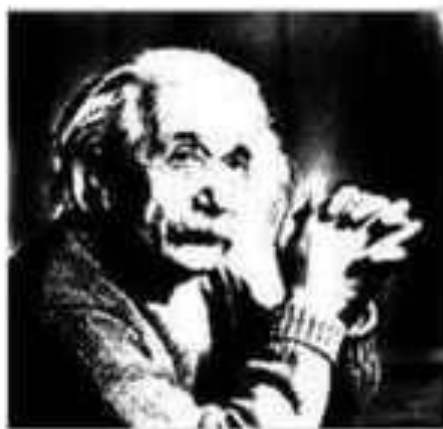


# POWER LAW TRANSFORMATION EXAMPLE

Gamma=10



Gamma=8



Gamma=6



## PIECEWISE LINEAR TRANSFORMATION FUNCTION

- There are three basic piecewise linear transformation functions.
  - ❖ Contrast stretching
  - ❖ Intensity level slicing
  - ❖ Bit plane slicing



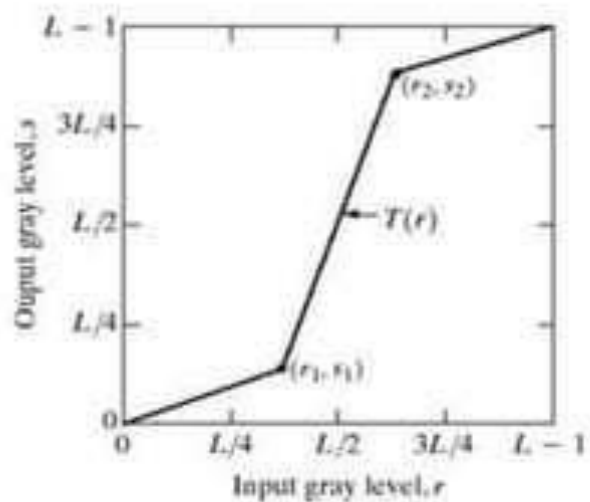
## CONTRAST STRETCHING

- It expands the range of intensity levels in an image.
- Contrast basically the difference between the intensity values of darker and brighter pixels.
- Contrast stretching is done in 3 ways,
  - ❖ Multiplying each input pixel intensity value with a constant scalar.
  - ❖ Using histogram equivalent.
  - ❖ Applying a transform which makes dark portion darker by assigning slope of  $< 1$  and bright portion brighter by assigning slope of  $> 1$ .



# CONTRAST STRETCHING EXAMPLE

Transformation function



Low contrast image



Contrast stretching image



## INTENSITY LEVEL SLICING

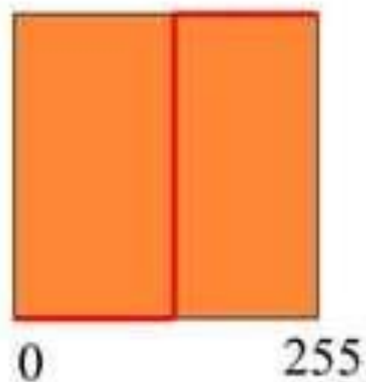
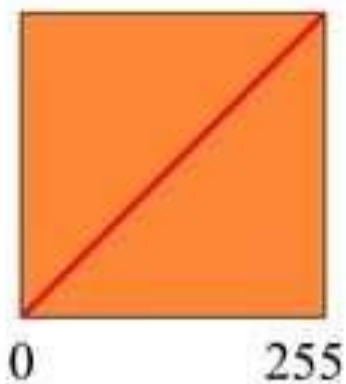
- Highlighting a specific range of gray levels in an image often is desired.
- There are several ways of doing level slicing, but most of them are variations of two basic themes.
- One approach is to display a high value for all gray levels in the range of interest.
- Another is a low value for all other gray levels.

# INTENSITY LEVEL SLICING EXAMPLE

Input image

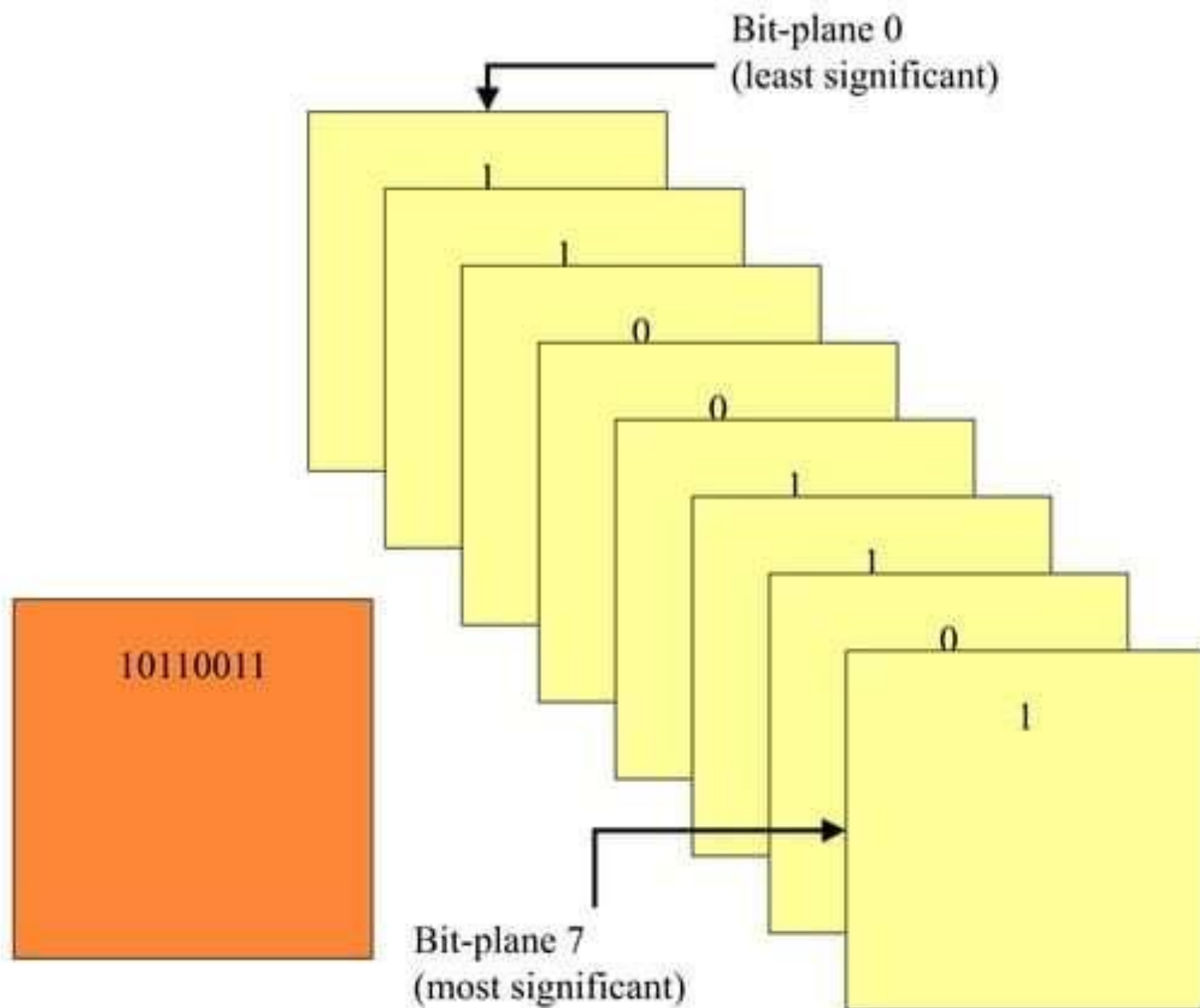


Output image



## BIT PLANE SLICING

- Instead of highlighting gray-level ranges, highlighting the contribution made to total image appearance by specific bits might be desired.
- Suppose that each pixel in an image is represented by 8 bits. Imagine that the image is composed of eight 1-bit planes, ranging from bit-plane 0, the least significant bit to bit-plane 7, the most significant bit.



# BIT PLANE SLICING EXAMPLE

Original image



Bit plane 7



Bit plane 6



Bit plane 4



Bit plane 1



THANK YOU !